Which reference organisms exercise which rapid panel targets? An open crosswalk of the CDC & FDA Antimicrobial Resistance Isolate Bank to FDA-cleared blood culture identification panels, with an external reference-method check of a U.S. benchmark

Fatima Abdulwahab [1,*]

[1] *Independent Researcher, Dallas, Texas, USA*

Before a laboratory reports results from a new test, U.S. federal rules require it to verify the test's performance, and characterized reference organisms are the usual material. The CDC & FDA Antimicrobial Resistance (AR) Isolate Bank is the main free U.S. source of such organisms, but it is organized by panel rather than by test, so a laboratory introducing a rapid blood culture identification panel cannot easily see which isolates exercise which resistance target. We built an open, versioned crosswalk from the Bank's public pages. For all 1,087 isolates in 39 panels we recorded the CDC gene list, the BioSample accession (975 isolates) and CDC's reference broth microdilution results, mapped 578 distinct CDC gene names to targets through the NCBI AMRFinderPlus Reference Gene Hierarchy, and aligned them with the resistance targets of 7 FDA-cleared panels taken from 9 FDA decision summaries. Of 825 isolates in organisms these panels test, 529 (64.1%) carry at least one target; every reported target except OXA-143-like is represented, from CTX-M (168 isolates in 17 panels) to *mecC* (2). Every panel's targets can be covered by two to five Bank isolates. We flag 169 isolates whose gene and growth results part ways, among them 14 that carry a target while testing not resistant (KPC with meropenem MICs of 0.25–2 mg/L, *mecA*-positive *S. aureus* susceptible to oxacillin and cefoxitin, a *vanA* operon lacking its regulators), and 38 carbapenem-resistant Enterobacterales and *Acinetobacter* without any panel carbapenemase. We then used the Bank as an external yardstick for a published U.S. genotype–phenotype benchmark. NCBI AMRFinderPlus reproduced CDC's target calls (Cohen's κ = 1.00 for eight targets, 0.93 for OXA-48-like, 0.98 for CTX-M), the benchmark's breakpoint rules reproduced CDC's interpretations for 99.8% of 2,851 results, and in 538 independent isolates the benchmark's gene-level estimates held wherever both could be estimated: a panel target predicted resistance to at least one carbapenem with PPV 96.6% and CTX-M predicted ceftriaxone resistance with PPV 100.0%. The crosswalk, code, provenance log and quality-control report are openly released.

Keywords: test verification, CLIA, reference isolates, AR Isolate Bank, rapid diagnostics, blood culture identification panel, carbapenemase, *mecA*, AMRFinderPlus

I. INTRODUCTION

Multiplex blood culture identification panels report the organism and a small set of resistance genes within hours of a positive blood culture [1, 2]. The genes were chosen because each is tied to a treatment decision: *mecA* and its divergent homologue *mecC* [3] for methicillin resistance in *S. aureus*; *vanA* and *vanB* for vancomycin resistance in enterococci [4]; the carbapenemases KPC, NDM, OXA-48-like, VIM and IMP; CTX-M, the dominant extended-spectrum β-lactamase (ESBL) family [5, 6]; and, on newer panels, *mcr*, SME and the OXA-23, -24/40 and -58 carbapenemases of *Acinetobacter*. Molecular rapid diagnostics on positive blood cultures have been associated with lower mortality and shorter time to effective therapy, particularly with antimicrobial stewardship [7, 8].

Before any of these results reaches a patient's chart, the laboratory must show the test works in its hands. Under the Clinical Laboratory Improvement Amendments, a laboratory that introduces an unmodified FDA-cleared test must, before reporting patient results, demonstrate performance comparable to the manufacturer's for accuracy, precision and reportable range [9]; CLSI has published guidance on verifying commercial microbial identification and susceptibility systems with characterized isolates [10]. For a resistance-gene target, accuracy can only be shown with organisms known to carry the target, organisms known not to, and, ideally, the difficult organisms in which the gene and the growth result disagree.

The CDC & FDA AR Isolate Bank was created in July 2015 to supply exactly such organisms [11]. Its isolates are characterized by reference broth microdilution and whole-genome sequencing and are offered free to approved institutions in curated panels [11, 12]. CDC reports that in its tenth year the Bank had grown from 3 to 34 panels and had shipped 500,000 isolates [13]. The Bank, however, is organized around resistance themes (carbapenemase diversity, borderline oxacillin susceptibility, new antibiotics) and around the drugs whose breakpoints the panels were built to test, not around diagnostic devices. A laboratory bringing up a BioFire BCID2 or a LIAISON PLEX panel has no list of which Bank isolates carry KPC, OXA-48-like or *mecC*, and no list of which ones will look discordant. It must open isolate pages one at a time.

*abdulwahabfatima43@gmail.com

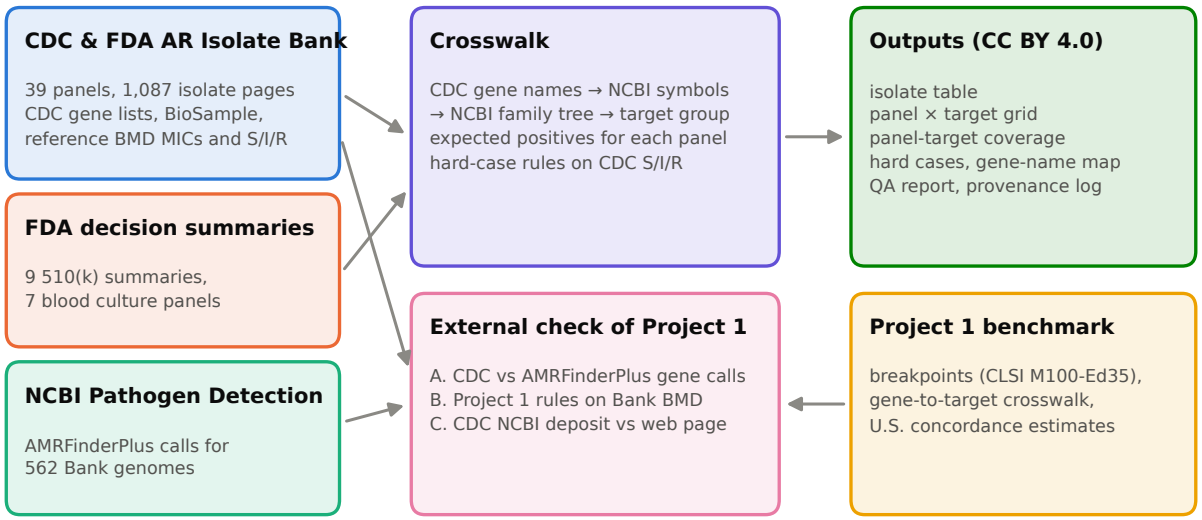


FIG. 1: Study design. Public CDC, FDA and NCBI sources feed two products: the crosswalk of Bank isolates to rapid panel targets, and an external check of the Project 1 benchmark. BMD, broth microdilution.

The same isolates also serve a second purpose. The author's U.S. Rapid Panel Resistance Marker Concordance Benchmark ("Project 1") measured how often each panel target predicts the susceptibility result reported for U.S. clinical isolates in NCBI Pathogen Detection [14]. Its phenotypes come from many submitting laboratories and platforms. The Bank's results come from one reference laboratory using the reference method, which makes the Bank a natural external check of that benchmark.

Here we describe the *CDC & FDA AR Isolate Bank: Rapid Panel Target Crosswalk*, version 1.0.0 [15]. It (i) captures every Bank isolate's gene list, BioSample accession and reference results exactly as CDC posts them; (ii) maps CDC's gene names to rapid panel targets through NCBI's gene family tree; (iii) states, for each of 7 FDA-cleared panels, which Bank isolates should be positive for each reported target; (iv) flags hard cases with explicit rules; and (v) uses the Bank to test the Project 1 benchmark. Section II describes the methods, Sec. III the results, Sec. IV their use and limits, and Sec. V our conclusions.

TABLE I: Data sources.

Source	Content used, version and terms
CDC & FDA AR Isolate Bank web site	All panel pages and 1,087 isolate pages (AR Bank number, species, BioSample, CDC gene list and calling method, reference broth microdilution MICs and S/I/R, page notes); accessed 24 Oct 2026; U.S. government work
FDA 510(k) decision summaries	9 summaries for 7 cleared blood culture panels with resistance targets; U.S. government works
NCBI Pathogen Detection	AMRFinderPlus genotypes for Bank BioSamples (versions 4.2.5, 4.2.7), streamed from the latest analysis of each organism group; no restrictions on use
AMRFinderPlus Reference Gene Hierarchy	Database 2026-08-07.1; gene-to-family tree used to assign targets
Project 1 benchmark v1.0.0	Breakpoint table (CLSI M100-Ed35, checked against FDA STIC), gene-to-target crosswalk, concordance table and isolate list

II. MATERIALS AND METHODS

A. Data sources

All inputs are public (Table I) and were downloaded on 24 October 2026. Every file was logged with its source URL, access date, size and SHA-256 fingerprint (1,188 provenance records).

1. Isolate Bank pages. We downloaded the Bank's *All Isolate Panels* page, every panel page it lists, the 5 custom panel pages linked from panel descriptions, and every isolate page reached from them (1,087 isolates). The downloader made one request at a time with a pause between requests and stored each page unchanged. From each isolate page we read the AR Bank number, species, study identifier, BioSample accession, specimen source, MLST type, the free-text note CDC sometimes places above the MIC table (for example a cefoxitin disk diffusion result), the MIC table (drug, MIC, CDC's S/I/R and footnotes), the interpretive standard CDC names for that page, the "Molecular Mechanisms of Resistance" table (category and gene) and CDC's note on how the genes were called. Panel membership was taken from the panel pages themselves.

TABLE II: Resistance targets of the FDA-cleared blood culture identification panels included.

Panel	FDA 510(k)	Resistance targets reported
BioFire BCID2	K193519 ; K243759	mecA/C, mecA/C and MREJ (MRSA), vanA/B, CTX-M, KPC, NDM, OXA-48-like, VIM, IMP, mcr-1
ePlex BCID-GP	K181663	mecA, mecC, vanA, vanB
ePlex BCID-GN	K182619 ; K213236	CTX-M, IMP, KPC, NDM, OXA, VIM
Verigene BC-GP	K122514	mecA, vanA, vanB
Verigene BC-GN	K132843	CTX-M, KPC, NDM, VIM, IMP, OXA
LIAISON PLEX GP	K243490	mecA/mecC, vanA, vanB
LIAISON PLEX GN	K243013	CTX-M, IMP, KPC, MCR, NDM, OXA, SME, VIM

2. Rapid panel targets. We took the resistance targets of each FDA-cleared blood culture panel from its 510(k) decision summary (Table II): BioFire BCID2 [16], ePlex BCID-GP and BCID-GN [17, 18], Verigene BC-GP and BC-GN [19, 20], and LIAISON PLEX Gram-Positive and Gram-Negative [21, 22]. Each reported target was translated into one or more target groups (for example, BCID2's "mecA/C" into *mecA* and *mecC*; ePlex BCID-GN's "OXA (OXA-23 and OXA-48 groups only)" into OXA-23-like and OXA-48-like). The Verigene BC-GN summary lists "OXA" without naming groups; we used the OXA-23 and OXA-48 groups used in its analytical studies, and the author confirmed this reading.

3. NCBI gene calls and gene hierarchy. For the external check we streamed the NCBI Pathogen Detection isolate tables (`latest_snps/AMR/*.amr.metadata.tsv`) for every organism group matching a Bank genus and kept rows whose BioSample is a Bank BioSample [23]. These rows carry the AMRFinderPlus genotype NCBI computed for each genome (versions 4.2.5, 4.2.7) [24]. The AMRFinderPlus Reference Gene Hierarchy (database 2026-08-07.1) was used to place genes in families [24].

4. Project 1 artifacts. From the Project 1 release we used, unchanged, its breakpoint table (CLSI M100, 35th edition [25], checked against the FDA Susceptibility Test Interpretive Criteria [26]), its AMRFinderPlus-symbol-to-target crosswalk, its concordance table and its list of benchmark BioSamples [14].

B. From CDC gene names to panel targets

CDC writes allele names without the *bla* prefix ("KPC-3", "OXA-232", "CTX-M-15"). Each of the 578 distinct names was matched to an NCBI symbol by trying, in order, the name itself, the name with *bla* added, NCBI synonyms and a case-insensitive match. When a name carried a one-letter bracket note ("OXA-48(c)"), the note was ignored after real gene names such as *tet*(A) had been tried. The NCBI symbol was then walked up the Reference Gene Hierarchy until it reached a node that defines a target group: *bla*KPC, *bla*NDM, *bla*VIM, *bla*IMP, *bla*CTX-M, the *bla*OXA-48, -23, -24, -58 and -143 families, *bla*SME, *mcr-1*, *mcr-2*, *mcr-3*, *mecA*, *mecC*, *vanA* and *vanB*. Three rulings were made by hand and recorded in the released gene map: "blaZ-mecC" (the *blaZ* carried on the *mecC* element), which CDC lists for the two isolates of its custom "*S. aureus* mecC+" panel, was counted as *mecC*; "vanHAX (lacks VanRS)", which CDC describes on the isolate page as a vancomycin-variable enterococcus, was counted as *vanA*; and the generic entry "MBL" was left unmapped because no enzyme is named.

C. Expected positives

An isolate was marked as an expected positive for a panel's reported target when it carried any target group the target covers and its organism is one for which the panel reports that target: *mecA*/*mecC* in staphylococci, *vanA*/*vanB* in enterococci, and Gram-negative targets in Enterobacterales, *Pseudomonas* and *Acinetobacter*. BCID2's MRSA result additionally requires *S. aureus*; the SCC*mec* right-extremity junction (MREJ) it detects cannot be read from a gene list, so every *mecA*- or *mecC*-positive *S. aureus* was assumed to be an expected MRSA positive. Fungal, gonococcal, *Clostridioides difficile* and enteric-pathogen panels were listed but not mapped, because these panels do not report resistance genes for those organisms.

D. Hard cases

Hard cases were defined on CDC's own posted interpretation, treating intermediate (I) and susceptible-dose-dependent as not resistant, as in Project 1. Imipenem was not used for *Proteus*, *Morganella* and *Providencia*, whose imipenem MICs are intrinsically elevated [25]. The rules were: (1) a target gene is present and none of the matching drugs is resistant (carbapenems for carbapenemases; ceftriaxone, cefotaxime or ceftazidime for CTX-M; oxacillin, cefoxitin MIC or, where CDC posts one, cefoxitin disk for *mecA*/*mecC*; vancomycin for *vanA*/*vanB*; colistin for *mcr*); (2) an Enterobacterales or *Acinetobacter* isolate is resistant to a carbapenem without any panel carbapenemase; (3) an Enterobacterales isolate is resistant to a third-generation cephalosporin without CTX-M or a carbapenemase; (4) a staphylococcus is resistant to oxacillin or cefoxitin without *mecA*/*mecC*; and (5) an enterococcus is resistant to vancomycin without *vanA*/*vanB*. Two patterns in which a negative panel result is the correct answer were kept as separate categories: carbapenem-resistant *P. aeruginosa* without a carbapenemase, usually caused by loss of the OprD porin, efflux and chromosomal AmpC [27], and enterococci whose vancomycin resistance comes from intrinsic *vanC* [28]. For rule 2 we classified the other mechanisms CDC lists, using NCBI's drug subclass for each gene: a carbapene-

mase not reported by any panel; an intrinsic OXA-51-like enzyme in *Acinetobacter*; a truncated porin with an ESBL or AmpC; an ESBL or AmpC alone; a truncated porin alone; or nothing listed.

E. Illustrative verification sets

To show how the crosswalk can be queried, we computed for each panel the smallest set of Bank isolates whose expected positives cover all of the panel's reported targets, by a deterministic greedy rule: repeatedly take the isolate that covers the most uncovered targets, preferring isolates that are not hard cases and then the lowest AR Bank number, using only isolates in panels on the Bank's main list. This is an example of a query, not a recommended verification protocol.

F. External check of the Project 1 benchmark

The check used Bank isolates in Project 1's organisms (*S. aureus*, *E. faecalis*, *E. faecium* and Enterobacterales) that have a BioSample.

1. Check A, gene calls. For Project 1's ten targets, we compared the targets in CDC's gene list with the targets in NCBI's AMRFinderPlus genotype for the same BioSample, mapped through the Project 1 crosswalk. Agreement is reported as counts and Cohen's κ [29]. Because some of CDC's gene calls used a database that includes NCBI's, the comparison was repeated in the isolates CDC called with ResFinder or ARG-ANNOT alone [30, 31].

2. Check B, gene versus reference result. CDC's MICs were reinterpreted with Project 1's breakpoint table and censoring rules, and every Project 1 analysis (target, drug and pooled organism group, with intermediate as not resistant) was recomputed with AMRFinderPlus target calls as the gene result. Isolates that are also in the Project 1 benchmark were excluded, so the Bank estimates come from isolates Project 1 never saw. Positive predictive value (PPV), negative predictive value, sensitivity and specificity are reported with Wilson 95% intervals [32] next to Project 1's estimates. Check B used MICs only, as in Project 1; CDC's cefoxitin disk results enter the hard-case rules but not this comparison.

3. Check C, consistency of CDC's records. For Bank isolates also in the Project 1 benchmark, Project 1's category for each drug (from the antibiogram deposited in NCBI) was compared with the reinterpreted category from CDC's web page.

G. Reproducibility and quality control

The pipeline is a set of numbered Python scripts (pandas [33]; figures with matplotlib and seaborn [34, 35]). Every number in this manuscript is read from files the pipeline writes. A quality-control report lists row counts, unmatched gene names, every interpretation difference and named spot checks. Before release, the saved pages were re-parsed with an independently written reader, which matched all 20,546 MIC rows and 8,298 gene rows; all saved pages matched their recorded fingerprints; spot checks were compared with the live CDC pages; and a clean rebuild from the archived inputs reproduced every released table byte for byte.

TABLE III: Isolate Bank collection by organism group. *In scope*: organisms for which a cleared panel reports resistance genes. *Target*: isolates carrying at least one rapid panel target. *Hard case*: isolates in any hard-case category, including the two expected-negative categories.

Organism group	Isolates	BioSample	Gene list	In scope	Target	Hard case
Enterobacterales	427	422	408	yes	289	81
Pseudomonas	140	137	131	yes	50	69
Acinetobacter	68	68	67	yes	54	6
S. aureus	129	128	126	yes	107	4
Other staphylococci	9	9	9	yes	8	0
Enterococcus	52	51	50	yes	21	9
Neisseria	94	52	19	no	0	0
Fungi	126	68	44	no	0	0
Other bacteria	42	40	8	no	0	0
Total	1,087	975	862	825	529	169

H. Ethics

The study used only public, de-identified data on bacterial and fungal isolates. No human subjects were involved and no isolates were handled.

III. RESULTS

A. The collection

The Bank listed 34 panels on its main page, and 5 more custom panels were linked from their descriptions, for 39 panels and 1,087 unique isolates (Table III, Fig. 1). Of these, 975 list a BioSample accession and 862 list resistance genes. In all, 825 isolates belong to organisms that the cleared panels report resistance genes for: 427 Enterobacterales, 140 *Pseudomonas*, 129 *S. aureus*, 68 *Acinetobacter*, 52 enterococci and 9 other staphylococci. The rest are fungi (126), *Neisseria* (94) and other bacteria (42), mostly *C. difficile* and *Campylobacter*.

CDC's pages are not uniform (Fig. 2). Genes were called with ResFinder using its June 2016 database for 289 isolates, ARG-ANNOT and ResFinder in 2018 for 40, ResFinder 3.0 for 30, and a combined ResFinder, ARG-ANNOT and NCBI database accessed in 2021 or 2022 for 362; 141 gene lists name no method. Interpretations cite CLSI M100 2025 for 783 isolates, the 2026 edition for 86, 2024 for 60 and 2020 for 1; fungal and fastidious-organism pages cite their own standards, and 103 pages name none. In addition, 802 isolate–drug interpretations on 494 isolates are footnoted as based on FDA breakpoints. Of the 578 distinct gene names, 478 matched an NCBI symbol and 66 mapped to a panel target group; the 100 unmatched names are mostly point mutations, efflux genes, porins and fungal targets that no panel reports.

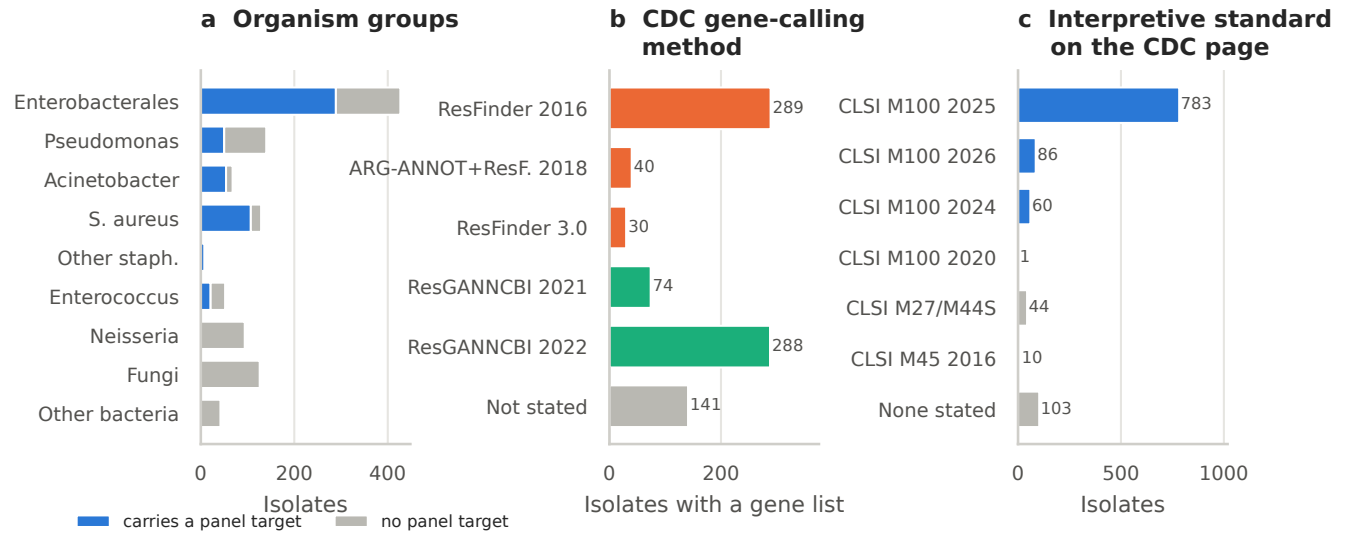


FIG. 2: The Isolate Bank collection. (a) Isolates by organism group and whether they carry a rapid panel target. (b) Method CDC states it used to call resistance genes; ResGANNCBI is CDC's combined ResFinder, ARG-ANNOT and NCBI database. (c) Interpretive standard named on the isolate page.

B. Target coverage

Of the 825 in-scope isolates, 529 (64.1%) carry at least one panel target, and 97 carry two or more (up to 4). Table IV gives the number of isolates, panels and species for each target group, and Fig. 3 shows where they sit. CTX-M is the most widely represented target (168 isolates in 17 panels), followed by *mecA* (113), KPC (95, 16 panels), NDM (77), OXA-48-like (38) and VIM (36). The *Acinetobacter* targets reported only by the newer Gram-negative panels are present (OXA-23-like 28, OXA-24/40-like 18, OXA-58-like 3), as are SME (9) and *mcr* genes (*mcr-1* 7, *mcr-2* 1, *mcr-3* 2). *mecC* is represented by two isolates (in the borderline-oxacillin panel and its "mecC+" custom panel), and OXA-143-like, which LIAISON PLEX Gram-Negative reports, by none.

The panels designed around carbapenemases cover the Gram-negative targets most densely: Gram Negative Carbapenemase Detection (CarbaNP) holds ten target groups, and Enterobacterales Carbapenemase Diversity (CRE) and the antibiotic verification panels for cefepime/zidebactam, plazomicin, imipenem/relebactam and meropenem/vaborbactam each hold four to seven. *mecA* sits mainly in the staphylococcal panels (borderline oxacillin susceptibility, difficult-to-detect *mecA*, oxazolidinone-resistant staphylococci and vancomycin-intermediate *S. aureus*), with further isolates in the delafloxacin and aminoglycoside/tetracycline panels; *vanA* and *vanB* sit mostly in the VRE panel; and *mcr* genes in the panel of isolates with new or novel resistance.

Fig. 4 sets the target groups each cleared panel reports against the number of Bank isolates that carry them, and the released coverage table gives, for each reported target, the expected-positive isolates and their panels. Every reported target of every panel has at least one expected-positive Bank isolate except OXA-143-like within LIAISON PLEX's OXA result, which its OXA-23, OXA-24/40, OXA-48 and OXA-58 groups cover. The greedy query (Table V) found that 5 isolates cover all 10 BCID2 resistance targets and 4 cover all 8 LIAISON PLEX Gram-Negative targets; across the seven panels, between 2 and 5 isolates suffice. Table V also lists, for each panel, the expected positives that test not resistant, which a laboratory may want to include deliberately.

C. Hard cases

In total, 169 isolates fall into at least one hard-case category (Table VI, Fig. 5); 75 of them are in the two categories where a negative panel result is expected and 94 in the others.

1. Target present, drug not resistant. Table VII lists the 14 isolates that carry a panel target while testing not resistant to every matching drug. Six are KPC producers (KPC-2, -3 or -4) in *E. cloacae*, *K. oxytoca*, *E. coli* and *P. mirabilis*, with meropenem MICs of 0.25 to 2 mg/L and imipenem and ertapenem MICs mostly intermediate. Two carry VIM: VIM-1 in *E. cloacae* (meropenem 1 mg/L) and VIM-11 in *P. aeruginosa* (all carbapenems at 4 mg/L, intermediate). One *Serratia marcescens* carries SME with every carbapenem at the lowest dilution tested. Three *S. aureus* carry *mecA* with oxacillin MICs of 0.25 to 0.5 mg/L and cefoxitin MICs of 4 mg/L (susceptible); two of them were also susceptible by cefoxitin disk. These resemble oxacillin-susceptible MRSA [36]. CDC posts a cefoxitin

TABLE IV: Rapid panel target groups in the Isolate Bank. *Species* counts distinct genus–species names as CDC writes them.

Target group	Isolates	Bank panels	Species	Panels reporting it[a]	CDC allele names (examples)
mecA	113	6	3	B2, eP, VP, LP	mecA
mecC	2	2	1	B2, eP, LP	blaZ-mecC
vanA	11	3	3	B2, eP, VP, LP	VanA, vanHAX(lacks VanRs)
vanB	10	1	2	B2, eP, VP, LP	VanB
KPC	95	16	14	B2, eN, VN, LN	KPC-11, KPC-2, KPC-3, KPC-4, KPC-45, KPC-5, +1 more
NDM	77	15	12	B2, eN, VN, LN	NDM, NDM-1, NDM-5, NDM-6, NDM-7
OXA-48-like	38	12	7	B2, eN, VN, LN	OXA-181, OXA-232, OXA-244, OXA-252, OXA-48, OXA-48(c)
VIM	36	11	4	B2, eN, VN, LN	VIM, VIM-1, VIM-11, VIM-2, VIM-27, VIM-4, +2 more
IMP	26	8	7	B2, eN, VN, LN	IMP-1, IMP-13, IMP-14, IMP-15, IMP-18, IMP-26, +7 more
CTX-M	168	17	12	B2, eN, VN, LN	CTX-M-1, CTX-M-124, CTX-M-14, CTX-M-14b, CTX-M-15, CTX-M-2, +6 more
OXA-23-like	28	4	1	eN, VN, LN	OXA-23
OXA-24/40-like	18	3	1	LN	OXA-24, OXA-72
OXA-58-like	3	2	1	LN	OXA-58
SME	9	3	1	LN	SME, SME-3, SME-4
mcr-1	7	1	3	B2, LN	mcr-1
mcr-2	1	1	1	LN	mcr-2
mcr-3	2	1	2	LN	mcr-3

[a]B2, BioFire BCID2; eP/eN, ePlex BCID-GP/GN; VP/VN, Verigene BC-GP/GN; LP/LN, LIAISON PLEX Gram-Positive/Gram-Negative. OXA-143-like is reported by LIAISON PLEX GN but no Bank isolate carries it. The MRSA result of BCID2 additionally requires the SCC*mec* right-extremity junction (MREJ), which a gene list cannot show.

TABLE V: Illustrative query: for each panel, the smallest set of Bank isolates (from panels on the Bank's main list) whose CDC gene lists cover every resistance target the panel reports, chosen by a deterministic greedy rule; and the expected-positive isolates whose matching drug tests not resistant. This shows how the crosswalk can be used; it is not a recommended verification protocol, and laboratories should follow the manufacturer's instructions and CLSI M52 [10].

Panel	Targets	Isolates	Smallest covering set: AR Bank number (targets exercised)	Target present, drug not resistant (AR Bank)
BioFire BCID2	10	5	1153 (CTX-M, KPC, NDM, OXA-48-like); 0215 (mecA/C, mecA/C and MREJ (MRSA)); 1159 (VIM, IMP); 0346 (mcr-1); 0571 (vanA/B)	0032, 0050, 0147, 0156, 0451, 0464, 0500, 0541, +5
ePlex BCID-GP	4	4	0215 (mecA); 0571 (vanA); 0782 (vanB); 1306 (mecC)	0464, 0500, 0804, 0998, 1003
ePlex BCID-GN	6	2	1153 (CTX-M, KPC, NDM, OXA); 1159 (IMP, VIM)	0032, 0050, 0147, 0156, 0451, 0541, 0611, 0834
Verigene BC-GP	3	3	0215 (mecA); 0571 (vanA); 0782 (vanB)	0464, 0500, 0804, 0998, 1003
Verigene BC-GN	6	2	1153 (CTX-M, KPC, NDM, OXA); 1159 (VIM, IMP)	0032, 0050, 0147, 0156, 0451, 0541, 0611, 0834
LIAISON PLEX GP	3	3	0215 (mecA/mecC); 0571 (vanA); 0782 (vanB)	0464, 0500, 0804, 0998, 1003
LIAISON PLEX GN	8	4	1153 (CTX-M, KPC, NDM, OXA); 1159 (IMP, VIM); 0091 (SME); 0346 (MCR)	0027, 0032, 0050, 0147, 0156, 0451, 0541, 0611, +1

disk result for 30 *mecA*-positive *S. aureus*, of which 28 are disk resistant; four of these (0977, 0981, 0991, 0992) are susceptible by oxacillin and cefoxitin MIC, so the disk detects their resistance and they are not counted as hard cases. Among enterococci, the vancomycin-variable *E. faecium* 0500 carries the *vanA* operon without its regulators and tests vancomycin susceptible (1 mg/L) [37, 38], and *E. faecalis* 0804 carries *vanB* with an intermediate vancomycin MIC of 16 mg/L.

2. Meropenem MICs of carbapenemase producers. Among Enterobacterales with a meropenem result (Fig. 6), 1 of 49 NDM producers and 1 of 9 SME producers tested not resistant, compared with 12 of 79 KPC, 7 of 23 OXA-48-like, 4 of 7 VIM and 7 of 10 IMP producers. No OXA-48-like producer was meropenem susceptible; all 7 not-resistant OXA-48-like isolates sat at the intermediate MIC of 2 mg/L, consistent with the weak carbapenemase activity of this family [39]. Most of these isolates were resistant to another carbapenem; only the isolates in Table VII were not resistant to any.

3. Resistant without a target. Of the 38 carbapenem-resistant Enterobacterales and *Acinetobacter* without a panel carbapenemase, the largest group (16) carry a truncated porin together with an ESBL or AmpC, 8 an ESBL or AmpC alone and 2 a truncated porin alone (Fig. 5b). Another 4 carry a carbapenemase that no cleared panel reports (NMC-A, of the IMI family, in two *Enterobacter*; OXA-237, of the OXA-134 family, in two *A. baumannii*), 3 *A. baumannii* carry only their intrinsic OXA-51-like enzyme, and for 5 CDC lists no β-lactam mechanism.

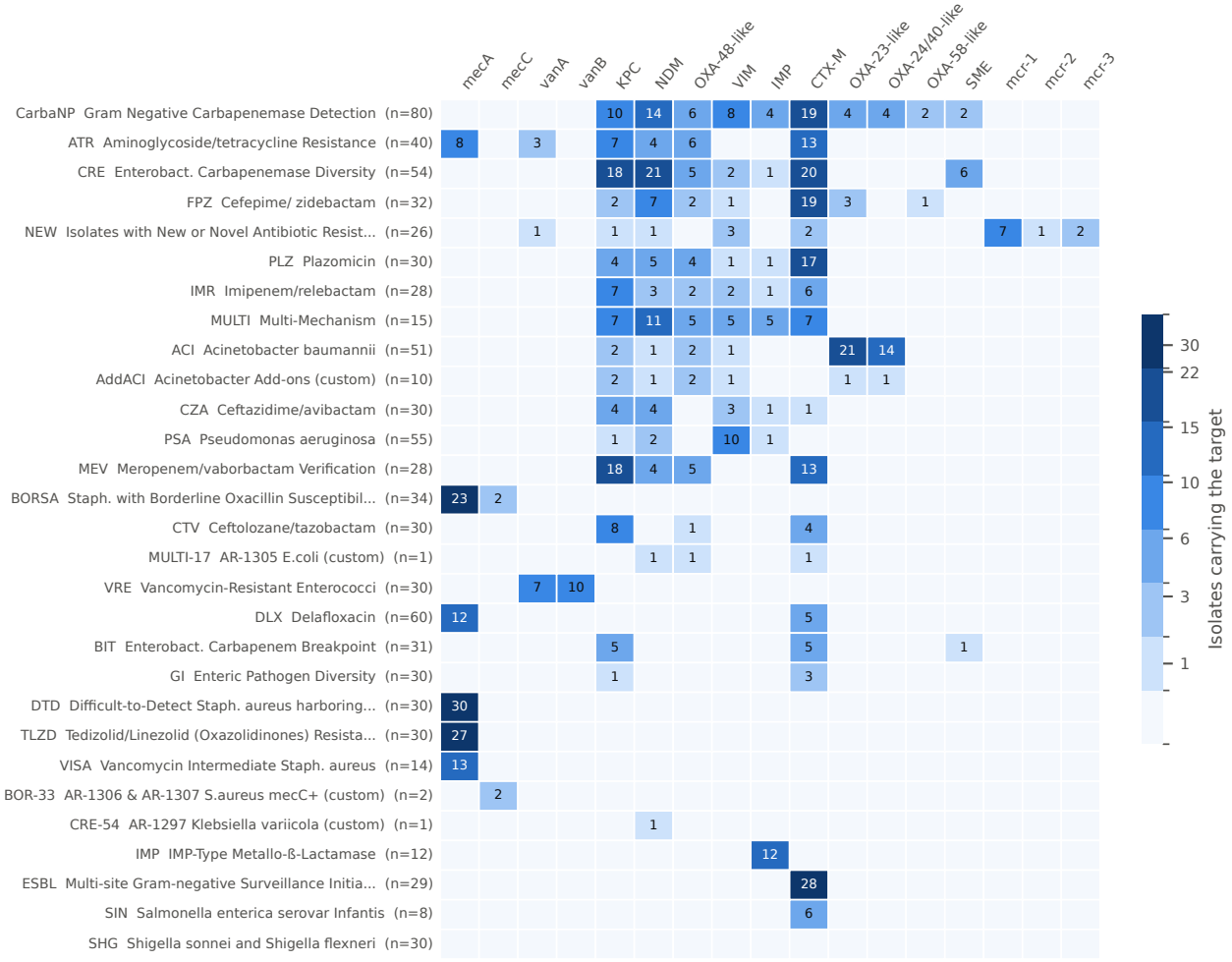


FIG. 3: Isolate Bank panels by rapid panel target. Each cell is the number of isolates in the panel whose CDC gene list includes the target; blank cells are zero. Panels without organisms that the cleared panels test are omitted. Panel codes are CDC's; "custom" panels are linked from other panels' descriptions. An isolate can belong to several panels.

A further 58 Enterobacterales are resistant to a third-generation cephalosporin without CTX-M or a carbapenemase, typically with SHV or TEM ESBLs or plasmid or chromosomal AmpC enzymes [40, 6]. One *S. aureus* is resistant to oxacillin and cefoxitin without *mecA* or *mecC*. Of the 140 *P. aeruginosa* isolates, 50 carry a panel carbapenemase and 68 are carbapenem resistant without one; for these a negative panel result is the correct answer [27]. Seven *E. gallinarum* and *E. casseliflavus* are vancomycin resistant through intrinsic *vanC* [28].

D. External check of the Project 1 benchmark

Of 593 Bank isolates in Project 1's organisms with a BioSample, 562 had AMRFinderPlus results in NCBI Pathogen Detection and 31 did not. Of these, 24 were also in the Project 1 benchmark; all 24 had been deposited in NCBI by CDC itself. They were excluded from check B, leaving 538 independent isolates.

1. Check A. CDC's gene lists and AMRFinderPlus agreed on every isolate for *mecA*, *mecC*, *vanA*, *vanB*, KPC, NDM, VIM and IMP (Table VIII, Fig. 7). OXA-48-like differed for 4 isolates, all *K. pneumoniae* with OXA-232 listed by CDC and absent from NCBI's analysis (κ = 0.93), and CTX-M for 2 in each direction (κ = 0.98). OXA-232 is carried on a small (6.1 kb) ColE-type plasmid [41], which may be missing from a public assembly while present in the sequencing reads; for three of the four isolates CDC states that its calls combined an assembly-based search with SRST2, which searches the reads directly. NCBI's analysis also confirmed *mecC* in both isolates CDC lists only as "blaZ-mecC". Agreement did not depend on CDC's method: 3 of 198 positive calls differed among isolates CDC called without NCBI's database, and 5 of 199 among those called with it.

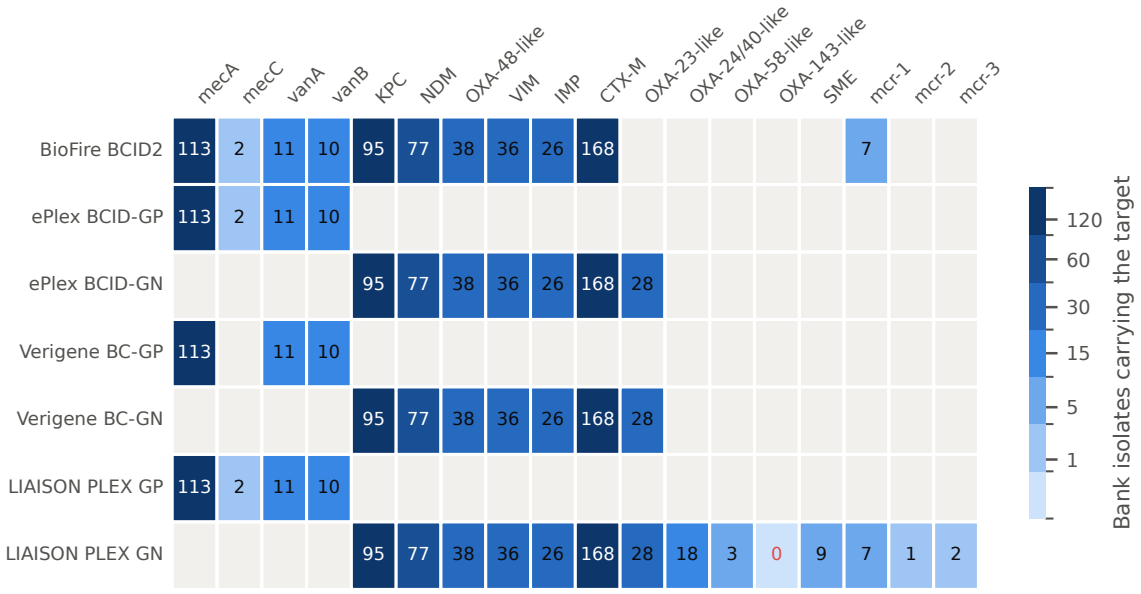


FIG. 4: Rapid panel target groups (columns) reported by each FDA-cleared panel (rows), with the number of Isolate Bank isolates whose CDC gene list includes the target group. Gray cells: the panel does not report that target group. Several groups are reported together as one result (for example BCID2's "mecA/C" and "vanA/B", or ePlex BCID-GN's "OXA"). The red 0 marks OXA-143-like, reported by LIAISON PLEX Gram-Negative but carried by no Bank isolate.

TABLE VI: Hard cases, by category, on CDC's posted interpretation (intermediate counted as not resistant). An isolate can fall in more than one category.

Category	Isolates	Breakdown	What a laboratory sees
Target gene present, matching drug(s) not resistant	14	KPC 6, mecA 3, VIM 2, SME 1, vanA 1, vanB 1	Panel positive, AST not resistant
Carbapenem resistant, no panel carbapenemase (Enterobacterales, *Acinetobacter*)	38	Porin truncation with ESBL or AmpC 16; ESBL or AmpC only 8; No mechanism listed by CDC 5; Carbapenemase not on any panel 4; Intrinsic OXA-51-like (Acinetobacter) 3; Porin truncation only 2	Panel negative, AST resistant
Third-generation cephalosporin resistant, no CTX-M or carbapenemase	58	Other ESBL (SHV, TEM), plasmid or chromosomal AmpC	CTX-M negative, AST resistant
Oxacillin or cefoxitin resistant, no *mecA*/*mecC*	1	No mechanism listed	Panel negative, AST resistant
Carbapenem-resistant *P. aeruginosa*, no carbapenemase	68	Chromosomal (OprD, efflux, AmpC)	Expected negative
Vancomycin resistant by intrinsic *vanC*	7	*E. gallinarum*, *E. casseliflavus*	Expected negative

2. Check B. Project 1's breakpoint rules reproduced CDC's posted category for 99.8% of 2,851 comparable isolate–drug results; the 6 differences (Table X) concern Morganellaceae meropenem and ceftazidime MICs at the breakpoint, an *Enterobacter* ceftazidime MIC of 8 mg/L, and one *mecA*-positive *S. aureus* with an oxacillin MIC of 2 mg/L that CDC reports as resistant, as CLSI directs for *mecA*-positive isolates [25].

Table IX and Fig. 8 set the Bank estimates beside Project 1's. When a target gene was present, the share of isolates testing resistant was similar to Project 1 for the high-PPV targets: KPC predicted resistance to any carbapenem in 60/64 (93.8%; Project 1 98.0%), OXA-48-like in 26/26 (100.0%), the five panel carbapenemases together in 142/147 (96.6%; Project 1 97.8%), and CTX-M predicted ceftriaxone resistance in 146/146 (100.0%; Project 1 97.7%). The drug-level pattern also held: carbapenemase producers were less often resistant to meropenem (PPV 84.4%) than to any carbapenem, and OXA-48-like producers least of all (19/26 meropenem resistant). For CTX-M and ceftazidime the Bank PPV was higher than Project 1's (87.6% versus 57.0%).

The measures that depend on what else is in the collection differed, as expected. Among *S. aureus*, *mecA*-positive isolates were often oxacillin susceptible by MIC (36 of them), so specificity for oxacillin was 39.0% against Project 1's 98.5%; with cefoxitin added, PPV was 94.2% and specificity 77.8%. These are the Bank's deliberately

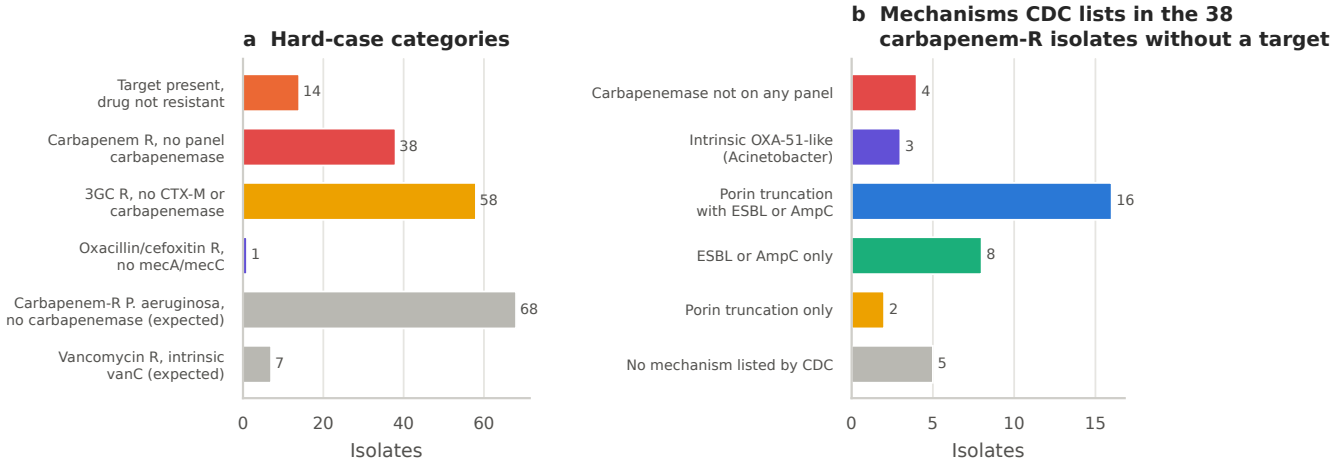


FIG. 5: Hard cases. (a) Isolates in each category; gray bars are the two categories in which a negative panel result is expected. (b) The mechanisms CDC lists for the 38 carbapenem-resistant Enterobacterales and *Acinetobacter* without a panel carbapenemase, classified by NCBI drug subclass.

TABLE VII: The 14 isolates that carry a rapid panel target while testing not resistant to every matching drug. Results are CDC's posted MICs (mg/L), disk diameters (mm) and interpretations.

AR Bank	Organism	Target	CDC gene	CDC reference results (MIC in mg/L or disk in mm, CDC S/I/R)
0032	*Enterobacter cloacae*	KPC	KPC-3	Meropenem=0.25 S; Imipenem=2 I; Ertapenem=1 I; Doripenem=0.5 S
0050	*Enterobacter cloacae*	KPC	KPC-4	Meropenem=2 I; Imipenem=2 I; Ertapenem=1 I; Doripenem=0.5 S
0147	*Klebsiella oxytoca*	KPC	KPC-3	Meropenem=0.5 S; Imipenem=2 I; Ertapenem=1 I; Doripenem=2 I
0156	*Proteus mirabilis*	KPC	KPC-2	Meropenem=0.25 S; Ertapenem=0.25 S; Doripenem=0.5 S
0451	*Escherichia coli*	KPC	KPC-3	Meropenem=0.5 S; Imipenem=2 I; Ertapenem=1 I; Doripenem=1 S
0541	*Escherichia coli*	KPC	KPC-3	Meropenem=1 S; Imipenem=2 I; Ertapenem=1 I; Doripenem=1 S
0027	*Serratia marcescens*	SME	SME	Meropenem=$\leq$0.12 S; Imipenem=$\leq$0.5 S; Ertapenem=$\leq$0.12 S; Doripenem=$\leq$0.12 S
0611	*Pseudomonas aeruginosa*	VIM	VIM-11	Meropenem=4 I; Imipenem=4 I; Doripenem=4 I
0834	*Enterobacter cloacae*	VIM	VIM-1	Meropenem=1 S; Imipenem=2 I; Ertapenem=0.5 S; Doripenem=2 I
0464	*Staphylococcus aureus*	mecA	mecA	Oxacillin=0.5 S; Cefoxitin=4 S
0998	*Staphylococcus aureus*	mecA	mecA	Oxacillin=$\leq$0.25 S; Cefoxitin=4 S; Cefoxitin disk=27 mm S
1003	*Staphylococcus aureus*	mecA	mecA	Oxacillin=0.5 S; Cefoxitin=4 S; Cefoxitin disk=26 mm S
0500	*Enterococcus faecium*	vanA	vanHAX(lacks VanRs)	Vancomycin=1 S
0804	*Enterococcus faecalis*	vanB	VanB	Vancomycin=16 I

TABLE VIII: Check A: rapid panel targets in CDC's gene lists and in NCBI AMRFinderPlus for the same BioSample. Right-hand columns restrict to isolates CDC called with ResFinder or ARG-ANNOT alone.

	All isolates with both calls					CDC method without NCBI database		
Target	*n*	Both	CDC only	NCBI only	κ	*n*	Both	Disagree
mecA	124	103	0	0	1.00	61	51	0
mecC	124	2	0	0	1.00	61	0	0
vanA	39	10	0	0	1.00	18	2	0
vanB	39	10	0	0	1.00	18	0	0
KPC	377	74	0	0	1.00	161	29	0
NDM	377	62	0	0	1.00	161	36	0
OXA-48-like	377	30	4	0	0.93	161	14	1
VIM	377	8	0	0	1.00	161	2	0
IMP	377	10	0	0	1.00	161	1	0
CTX-M	377	160	2	2	0.98	161	63	2

difficult staphylococci: its borderline oxacillin (BORSA) and difficult-to-detect *mecA* (DTD) panels. Sensitivity of each single carbapenemase was low because Bank isolates carry many different carbapenemases, but sensitivity of the five together for any-carbapenem resistance was 79.8% (Project 1 64.3%) and for meropenem 83.8% (Project 1 73.2%). CTX-M sensitivity for ceftriaxone resistance matched Project 1 (55.3% versus 56.5%), confirming that a negative CTX-M result is a weak predictor of ceftriaxone susceptibility.

3. Check C. For the 120 isolate–drug results on the 24 overlapping isolates, the category from CDC's NCBI antibiogram matched the category from CDC's web page in 97.5%, with 0 major or very major differences; the 3 differences were susceptible versus intermediate at a carbapenem breakpoint. Because both records come from CDC, this is a consistency check of CDC's two postings, not a comparison between laboratories.

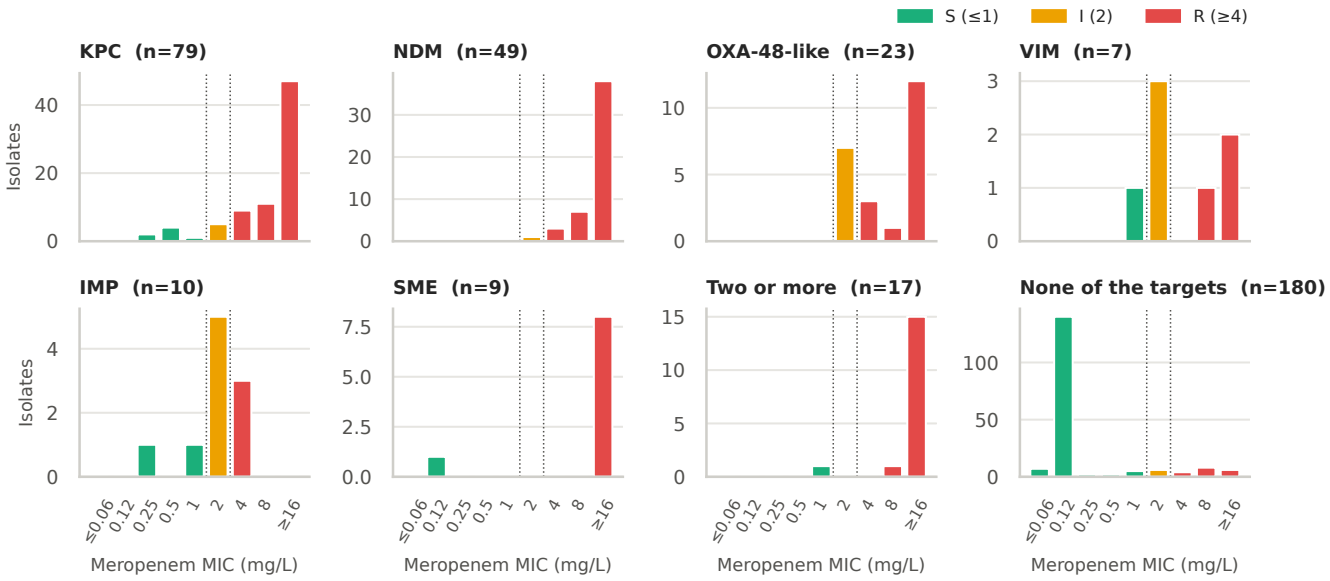


FIG. 6: CDC reference meropenem MICs for Enterobacterales, by the carbapenemase target the isolate carries. Colors give the CLSI category (susceptible $\leq$1, intermediate 2, resistant $\geq$4 mg/L); dotted lines mark the breakpoints. MICs recorded as $\leq$ or $>$ a value are shown at that value.

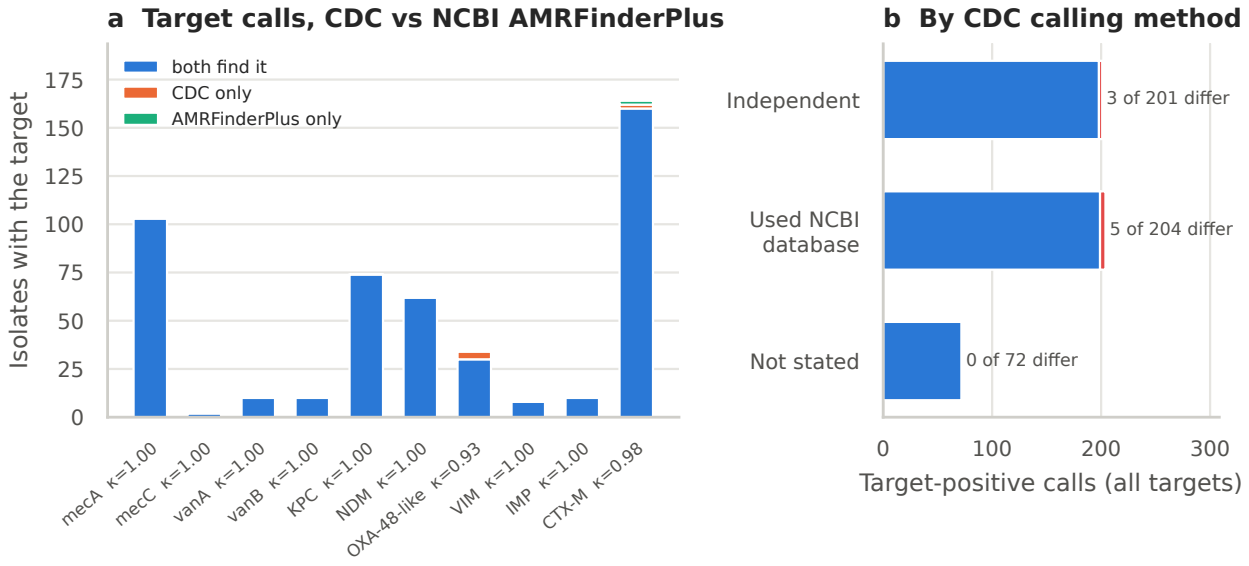


FIG. 7: Check A. (a) Isolates in which CDC, NCBI AMRFinderPlus, or both found each target, with Cohen's κ. (b) Disagreeing target-positive calls by CDC's gene-calling method; "independent" means CDC used ResFinder or ARG-ANNOT without NCBI's database.

IV. DISCUSSION

A. Principal findings

The Isolate Bank already contains reference organisms for nearly every resistance target reported by the seven FDA-cleared blood culture panels with resistance targets, and for most of the ways those targets can mislead. What it lacked was an index by target and by device. The crosswalk supplies it: each isolate's targets, the panels that should call it positive, its reference results, and whether it is a hard case. Two to five isolates cover every target of any one panel, and the hard-case list offers a further set of organisms that tests what a panel cannot know.

Used as an external yardstick, the Bank supports the Project 1 benchmark where the benchmark is meant to be read. Gene calls by two different pipelines agreed almost

TABLE IX: Check B: Project 1 rules applied to independent Bank isolates (Bank) beside Project 1's U.S. estimates (P1). *n* is Bank isolates with an interpretable result; PPV and sensitivity also give Bank counts. Enterobacterales exclude *Salmonella*; enterococci are *E. faecalis* and *E. faecium*. – , not estimable.

			PPV			Sensitivity			Specificity	
Target	Drug	*n*	Bank	Bank	P1	Bank	Bank	P1	Bank	P1
mecA	oxacillin	127	67/103	65.0%	97.7%	67/68	98.5%	100.0%	39.0%	98.5%
mecA	cefoxitin	127	97/103	94.2%	–	97/100	97.0%	0.0%	77.8%	–
mecA	oxacillin or cefoxitin	127	97/103	94.2%	97.7%	97/100	97.0%	99.5%	77.8%	98.5%
vanA	vancomycin	41	9/10	90.0%	89.7%	9/18	50.0%	89.7%	95.7%	96.7%
vanB	vancomycin	41	9/10	90.0%	–	9/18	50.0%	0.0%	95.7%	100.0%
KPC	meropenem	312	54/64	84.4%	90.5%	54/148	36.5%	55.3%	93.9%	99.5%
KPC	any carbapenem	312	60/64	93.8%	98.0%	60/178	33.7%	47.0%	97.0%	99.9%
NDM	meropenem	312	52/53	98.1%	97.6%	52/148	35.1%	12.2%	99.4%	100.0%
NDM	any carbapenem	312	53/53	100.0%	100.0%	53/178	29.8%	9.7%	100.0%	100.0%
OXA-48-like	meropenem	312	19/26	73.1%	76.3%	19/148	12.8%	8.8%	95.7%	99.7%
OXA-48-like	any carbapenem	312	26/26	100.0%	94.7%	26/178	14.6%	8.5%	100.0%	100.0%
VIM	meropenem	312	3/6	50.0%	75.0%	3/148	2.0%	0.9%	98.2%	100.0%
IMP	meropenem	312	1/3	33.3%	37.5%	1/148	0.7%	0.9%	98.8%	99.9%
CTX-M	ceftriaxone	350	146/146	100.0%	97.7%	146/264	55.3%	56.5%	100.0%	99.1%
CTX-M	ceftazidime	312	127/145	87.6%	57.0%	127/235	54.0%	42.7%	76.6%	86.7%
mecA or mecC	oxacillin or cefoxitin	127	99/105	94.3%	97.7%	99/100	99.0%	99.5%	77.8%	98.5%
vanA or vanB	vancomycin	41	18/20	90.0%	89.7%	18/18	100.0%	89.7%	91.3%	96.7%
Any of 5 carbapenemases	meropenem	312	124/147	84.4%	87.6%	124/148	83.8%	73.2%	86.0%	99.0%
Any of 5 carbapenemases	any carbapenem	312	142/147	96.6%	97.8%	142/178	79.8%	64.3%	96.3%	99.8%

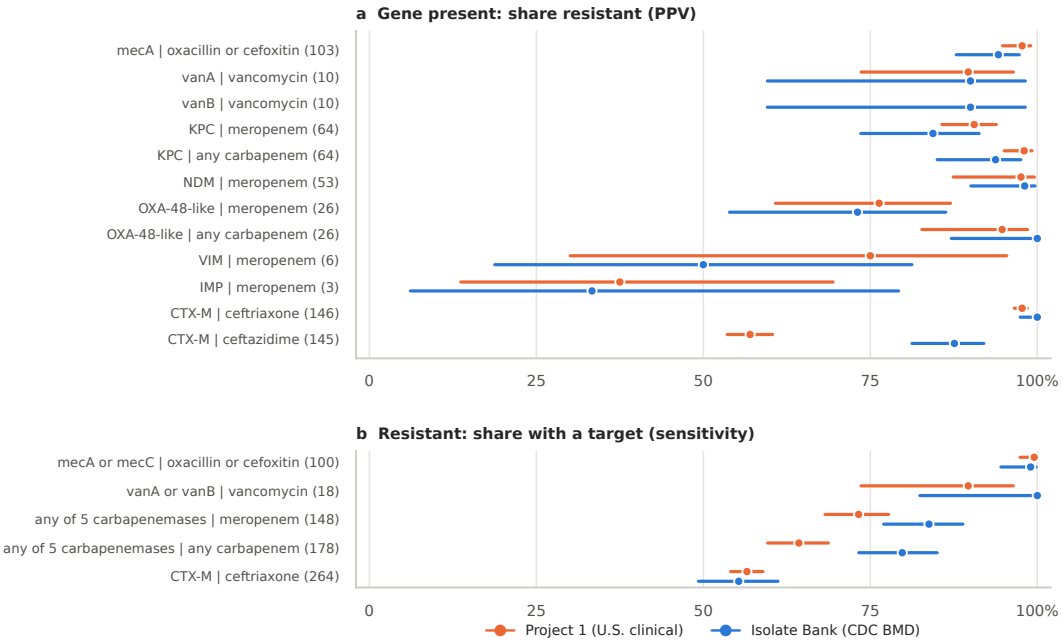


FIG. 8: Check B. (a) Share of gene-positive isolates that test resistant (PPV) and (b) share of resistant isolates that carry a target (sensitivity), Bank (blue) beside Project 1 (orange), with 95% Wilson intervals. Numbers in brackets are Bank denominators.

perfectly. The benchmark's breakpoint rules reproduced CDC's reference interpretations. And the quantities that describe the gene itself (how often a gene-positive isolate is resistant, and to which carbapenem) behaved the same way in reference broth microdilution data from one laboratory as in submitted results from many.

B. Using the crosswalk for verification

A laboratory verifying a panel can filter the released isolate table by the panel's `expected_positive` column to see every candidate positive, by `hard_case_types` to find challenge organisms, and by organism to find negative controls of the right species. The covering sets in

TABLE X: The 6 isolate–drug results where Project 1's breakpoint rules and CDC's posted category differ.

AR Bank	Organism group	Drug	MIC	CDC	Project 1 rules
0057	Morganella / Providencia	meropenem	2	R	I
0082	Morganella / Providencia	meropenem	4	I	R
0155	Morganella / Providencia	ceftazidime	8	R	I
0216	*Staphylococcus aureus*	oxacillin	2	R	S
1011	*Enterobacter cloacae* complex	ceftazidime	8	R	I
1113	Morganella / Providencia	meropenem	2	S	I

Table V show that coverage is cheap; the hard-case list is where the crosswalk adds most. A KPC-producing *E. coli* with a meropenem MIC of 0.5 mg/L, a *mecA*-positive *S. aureus* susceptible by oxacillin and cefoxitin MIC, or a vancomycin-susceptible *E. faecium* carrying *vanA* will each produce a positive panel result and a susceptible AST result. Including them in a verification shows the laboratory and its stewardship program what such a result looks like before it happens on a patient isolate, and tests whether the laboratory's reporting rules (for example, reporting *mecA*-positive *S. aureus* as oxacillin resistant [25]) are applied. Conversely, the carbapenem-resistant isolates without a panel carbapenemase, and especially the NMC-A, OXA-237 and porin-loss isolates, show the limits of a negative carbapenemase result [42, 43].

C. What the external check shows

The check separates measures that describe a gene from measures that describe a population. PPV for KPC, NDM and OXA-48-like against any carbapenem, and for CTX-M against ceftriaxone, had overlapping 95% ranges in the Bank and in Project 1, and the low meropenem PPV of OXA-48-like producers reappeared in reference broth microdilution data [39]. Specificity for *mecA* and single-gene sensitivities, in contrast, were lower in the Bank because the Bank was built to contain the exceptions: borderline and difficult-to-detect staphylococci, and carbapenem-resistant Enterobacterales with every mechanism CDC could collect. These differences are therefore not evidence against Project 1's estimates for U.S. clinical isolates; they show what those estimates look like in a challenge set. The higher CTX-M PPV for ceftazidime in the Bank than in Project 1 may reflect the Bank's many multi-mechanism isolates; we did not test this.

D. Limitations

The crosswalk records which targets an isolate carries, not whether a particular assay detects that allele; detection is defined by each manufacturer's instructions for use. The MRSA result of BCID2 depends on the MREJ junction, which a gene list cannot show, so its expected positives are an assumption. The Verigene BC-GN OXA groups are not named by FDA and were taken from its analytical studies. CDC's gene lists came from several methods between 2016 and 2022, and older calls can miss genes added to databases later; the gene-call comparison is only partly independent of NCBI, although agreement was as high in the independent subset. Hard cases rest on CDC's posted interpretations, which cite four CLSI M100 editions, while the external check uses a single breakpoint table. The Bank is not a patient population, so its prevalence-dependent measures should not be compared directly with clinical estimates. Some targets are represented by few isolates (*mecC* by two, OXA-58-like by three) and OXA-143-like by none. Thirty-one Bank isolates in Project 1's organisms had no NCBI AMRFinderPlus result and 112 isolates list no BioSample. The release reflects the Bank's pages on 24 October 2026; CDC adds isolates and revises MICs, so the crosswalk will need yearly refresh. No isolates were tested.

V. CONCLUSIONS

An open, versioned crosswalk turns the CDC & FDA AR Isolate Bank into a target-by-target, device-by-device catalog for verifying rapid blood culture panels: 529 isolates carry a panel target, every target but one is represented, a handful of isolates covers each panel, and 94 isolates are flagged as hard cases outside the expected-negative groups. Used as an external reference-method check, the Bank confirmed the gene calls, interpretation rules and gene-level estimates of a U.S. genotype–phenotype benchmark. Both resources are released for reuse and will be refreshed as the Bank grows.

DATA AND CODE AVAILABILITY

All data, code and documentation are available at `https://github.com/FatimaAbdulwahab/ar-isolate-bank-panel-crosswalk` [15]: the isolate-level crosswalk, the panel-by-target grid, panel target coverage, the hard-case list, the CDC gene-name map, the rapid panel target table with FDA sources, the external-check outputs, a codebook, a limitations file, the quality-control report and a provenance log with source URLs, access dates, sizes and SHA-256 fingerprints. The CDC pages, NCBI rows and FDA documents used are archived in the repository, and `run_all.sh` rebuilds every result from them. Data and documents are released under CC BY 4.0 and code under the MIT license. The Project 1 benchmark is at `https://github.com/FatimaAbdulwahab/us-rapid-panel-amr-concordance` [14].

AUTHOR CONTRIBUTIONS

F.A.: conceptualization, methodology, data curation, formal analysis, validation, visualization, writing (original draft, review and editing).

COMPETING INTERESTS

The author declares no competing interests and no relationship with CDC, FDA or any manufacturer of rapid diagnostic panels. This work is not affiliated with or

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USE OF AI TOOLS

Claude (Anthropic) assisted with code, data processing, figure generation and drafting of this manuscript. The author made the analytic decisions and rulings, verified the outputs against the CDC, FDA and NCBI records, and takes responsibility for the content.

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